

Ga (Ga I). Strongest classified transitions below 2 μm (NIST): 17 levels, 23 lines.



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$3d^{10}4s^2 4p^2 P^o_{3/2}$	1338.9938(5)	-	F=1: -1673.74	F=2: +1004.25		
$3d^{10}4s^2 4p^2 P^o_{3/2}$	191.5(9)	63.0(20)	F=0: -639.38	F=1: -510.88	F=2: -190.88	F=3: +446.62
$4s^2 5s^2 S_{1/2}$	1069.5(15)	-	F=1: -1336.88	F=2: +802.12		
$4s^2 4d^2 D_{3/2}$	-36.3(22)	5(4)	F=0: +141.88	F=1: +100.97	F=2: +23.77	F=3: -80.53
$4s^2 4d^2 D_{5/2}$	77.3(9)	5(4)	F=1: -402.12	F=2: -251.75	F=3: -22.24	F=4: +291.20

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle V \| er \| J' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation

Level caption: NIST energy and measured lifetime (where one exists).

Isotope shifts

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$